

1. Administrative & Study Identification

Full Study Title: A Phase II, Open-Label, Multicenter Study Evaluating the Safety and Efficacy of Neoadjuvant Treatment With Inavolisib Combinations in Patients With Untreated, Early-stage, PIK3CA-Mutated Breast Cancer **Short Title/Acronym:** A Study to Test Inavolisib Treatments in Participants With Early-Stage, PIK3CA-Mutated Breast Cancer **Protocol Number:** B045853 / 2024-518811-20-00 **NCT Number:** NCT07054190 **Posting ID:** 670581411340092875 **Trial Status:** Recruiting (Currently recruiting participants) **Sponsor/Company Name:** Hoffmann-La Roche / Roche **Investigational Product:** Inavolisib, Ribociclib, and Letrozole **Phase of Development:** Phase II **Medical Monitor:** Hoffmann-La Roche Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the safety and tolerability of neoadjuvant treatment with inavolisib combinations in patients with untreated, early-stage, PIK3CA-mutated breast cancer. **Secondary Objectives:** To evaluate the preliminary efficacy of the combinations by assessing the total pathological complete response (pCR), objective response rate (reduction of cancer), and changes in the cellular proliferation marker Ki-67.

2.2 Background & Rationale Early-stage breast cancer is confined to the breast and has not spread. In estrogen receptor (ER)-positive breast cancers, estrogen drives cancer cell growth, making hormone blockers the standard of care. However, a subset of patients has tumors harboring a *PIK3CA* mutation, which can alter the cancer cells in ways that confer resistance to standard endocrine therapies. This study is designed to test the investigational PI3K inhibitor inavolisib, administered in combination with standard treatments (ribociclib and letrozole), to see how safe and effective this combination is in shrinking the tumor prior to surgery (neoadjuvant setting).

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active experimental arms **Blinding:** Open-label **Randomization:** Multicenter, assigned to treatment arms (e.g., Arms A, B, C)

3.2 Planned Enrollment & Duration Target \$N\$: 60 participants **Number of Sites:** Multicenter (Global sites, including locations

in Canada and Germany) **Estimated Study Start:** 2024 / 2025
Estimated Primary Completion: [Date TBD pending full enrollment]

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Female, ≥ 18 years of age. 2. Histologically confirmed operable or inoperable invasive Stage II-III Breast Cancer (BC) according to the American Joint Committee on Cancer (AJCC) TNM staging classification. 3. Documented ER-positive and HER2-negative tumor in accordance with current ASCO/CAP guidelines. 4. Confirmed PIK3CA mutation and documented Ki-67 score $\geq 5\%$ as per local assessment. 5. Candidate for neoadjuvant treatment and considered appropriate for endocrine combination therapy, with a willingness to undergo breast surgery (mastectomy or breast-conserving surgery) after treatment (unless inoperable).

4.2 Exclusion Criteria 1. Stage IV (metastatic) BC, Inflammatory BC (cT4d), or Bilateral invasive BC. 2. Previous systemic or local treatment for the primary BC currently under investigation (including excisional biopsy or any other surgery of the primary tumor and/or axillary lymph nodes, including sentinel lymph node biopsy, radiotherapy, cytotoxic, and endocrine treatments). 3. History of ductal carcinoma in situ or lobular carcinoma in situ if they have received any systemic therapy for treatment or radiation therapy to the ipsilateral breast. 4. History of Type 1 diabetes, or Type 2 diabetes requiring ongoing systemic treatment at the time of study entry. 5. Pregnant or breastfeeding women.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: * Inavolisib: Taken once a day. * Letrozole (Trade Name: Femara, ATC Code: L02BG04): Taken once a day. * Ribociclib: Taken daily for 21 days followed by 7 days off (28-day cycle). * LHRH Agonist: Women who can get pregnant must take an ovulation-blocking medication starting at least 14 days before and continuing during the study. **Route of Administration:** Oral (Tablets to be swallowed). **Duration of Treatment:** Approximately 5 to 6 months prior to surgery.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care medications per investigator discretion. **Prohibited:** Concurrent investigational therapies, or any other systemic anti-cancer treatments outside of the study protocol.

6. Study Timeline & Visit Schedule

| Visit ID | Window (Days) | Key Activities/Procedures | | :---
| :--- | :--- | | **Screening** | Day -28 to 0 | Informed Consent, Medical History, Tumor Biopsy, Biomarker Testing (PIK3CA, ER/

HER2, Ki-67), LHRH agonist initiation (if applicable) | |
Baseline | Day 0 | Treatment assignment, First Dose | | **Treatment Phase** | Months 1-6 | Administration of Inavolisib + Letrozole + Ribociclib, Safety Labs, AE Monitoring | | **End of Treatment** | Month 5-6 | Final scans, Pre-surgical evaluation, Treatment completion | | **Surgery & Follow-up** | Post-treatment | Mastectomy or breast-conserving surgery, Pathological evaluation of residual tumor (pCR assessment) |

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Safety and Tolerability: Percentage of participants with Adverse Events (AEs) associated with the inavolisib combination treatment. **Measurement Method:** Clinical assessment and laboratory monitoring graded per standard toxicity criteria (e.g., CTCAE).

7.2 Secondary & Exploratory Endpoints * **Pathological Complete Response (pCR):** Percentage of participants with Total Pathological Complete Response (pCR) with no detectable cancer left in the breast tissue and lymph nodes after neoadjuvant treatment. * **Objective Response Rate (ORR):** Percentage of participants with Objective Response Rate (ORR), according to Modified Response Evaluation Criteria in Solid Tumors, Version 1.1 (mRECIST v1.1). * **Biomarker Outcomes:** Change from baseline in Ki-67 levels by Immunohistochemistry (IHC).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring for known unwanted effects, which may include nausea, diarrhea, rash, stomatitis (sore mouth), hyperglycemia (high blood sugar), fatigue, neutropenia, and hepatotoxicity. **Serious Adverse Events (SAEs):** Reported within 24 hours to the sponsor and Institutional Review Boards. **Data Monitoring Committee (DMC):** An independent or internal safety monitoring committee will regularly review unblinded safety data.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: * **Safety Set:** All participants who receive at least one dose of the investigational product. * **Efficacy Set / Intent-to-Treat (ITT):** All assigned/enrolled participants evaluated for pCR and ORR. **Sample Size Justification:** Target enrollment of 60 participants, sized appropriately for a Phase II oncology trial to detect meaningful safety signals and early efficacy (pCR) estimations. **Handling of Missing Data:** Standard oncology imputation protocols; participants lost to follow-up prior to surgery may be considered non-responders for pCR.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Routine onsite and remote monitoring to ensure protocol compliance and data integrity. **Audit Trail:** Secure Electronic Data Capture (EDC) systems with full audit trails for data clarification and resolution.

11. Study Identifiers & Governance

Primary ID: B045853 / 2024-518811-20-00 **Registry Number:**

NCT07054190 **Posting ID:** 670581411340092875 **Sponsor/Lead**

Organization: Hoffmann-La Roche / Roche **Study Title:** A Study to Test Inavolisib Treatments in Participants With Early-Stage, PIK3CA-Mutated Breast Cancer **Official Regulatory Title:** A Phase II, Open-Label, Multicenter Study Evaluating the Safety and Efficacy of Neoadjuvant Treatment With Inavolisib Combinations in Patients With Untreated, Early-stage, PIK3CA-Mutated Breast Cancer

12. Clinical Framework (Breast Cancer Specific)

Primary Condition: Early-Stage Breast Cancer (Stage II-III)

Diagnosis Threshold: ER-positive, HER2-negative, PIK3CA-mutated, Ki-67 $\geq 5\%$

Medical Methodology: Neoadjuvant Endocrine Combination Therapy **Optimization Target:** Maximal tumor reduction (Pathological Complete Response) prior to surgical resection

13. Study Procedures & Methodology

Management Pathway: Oral neoadjuvant systemic therapy followed by surgery **Education Protocol (HCP):** Site initiation visits and detailed dosing/toxicity management guidelines for PI3K and CDK4/6 inhibitors.

Education Protocol (Patient): Informed consent details regarding the investigational nature of inavolisib, stringent contraception requirements, and AE reporting. **Quality**

Audit Mechanism: Centralized laboratory verification of PIK3CA mutation and rigorous clinical monitoring.

14. Observation & Follow-Up Schedule

Total Duration: Approximately 6 months of active neoadjuvant treatment.

Onsite Visit Cadence: Pre-treatment baseline, regular interval monitoring throughout the 21-day/7-day off ribociclib cycles, and pre-operative evaluation. **Remote Monitoring**

Frequency: As per specific site monitoring plans. **Checklist**

Review Interval: Regular cycle reviews (every 4 weeks) for safety labs, fasting glucose, and tolerability.

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				